Code Alpha Internship

Submission of Task-1 (Sensor-Based Simulation) - Full Code Explanation

Project Title: IoT-Based Motion Detection Alarm System using Arduino and PIR Sensor

Code Explanation

1. Pin Declaration

```
int pirPin = 4;
int ledPin = 5;
int buzzer = 6;
int dt = 500;
```

This section defines the pins used in the project. **pirPin** stores the PIR motion sensor pin number. **ledPin** stores the LED pin number. **buzzer** stores the buzzer pin number. **dt** stores the delay time in milliseconds.

2. Setup Function

```
  void setup(){
  pinMode(pirPin,INPUT);
  pinMode(ledPin,OUTPUT);
  pinMode(buzzer,OUTPUT);
  Serial.begin(9600);
}
```

The setup() function runs only once when the Arduino starts. **pinMode()** is used to configure the PIR sensor as INPUT and LED and buzzer as OUTPUT. **Serial.begin(9600)** starts serial communication to display messages on the Serial Monitor.

3. Loop Function

```
 void loop()
```

The loop() function runs continuously and checks the PIR sensor repeatedly.

4. Reading PIR Sensor Data

```
int motion = digitalRead(pirPin);
```

This line reads the output from the PIR sensor. If the output is HIGH, motion is detected. If the output is LOW, no motion is detected.

5. Motion Detection Condition

```
  if(motion == HIGH)
```

This condition checks whether motion has been detected. When motion is detected: The LED turns ON. The buzzer alarm activates. A message is displayed on the Serial Monitor.

6. LED Activation

```
digitalWrite(ledPin,HIGH);
```

This line turns ON the LED when motion is detected.

7. Displaying Message

```
Serial.println("Motion Detected!");
```

This displays the message “Motion Detected!” on the Serial Monitor.

8. Increasing Siren Frequency

```
for(int freq = 700;freq <= 2000; freq += 5)
```

This loop gradually increases the buzzer frequency from 700 Hz to 2000 Hz to create a rising siren effect.

9. Generating Sound

```
tone(buzzer,freq);
```

This function generates sound on the buzzer using the current frequency value.

10. Delay Function

```
delay(2);
```

This adds a small delay to make the siren sound smooth.

11.Decreasing Siren Frequency

```
for(int freq = 2000; freq >= 700; freq -= 5)
```

This loop gradually decreases the buzzer frequency from 2000 Hz to 700 Hz to create a falling siren effect.

12.No Motion Condition

```
else
```

This block executes when no motion is detected. The LED turns OFF. The buzzer stops. A “No Motion” message is displayed.

13.Turning OFF LED

```
digitalWrite(ledPin,LOW);
```

This line turns OFF the LED.

14.Stopping the Buzzer

```
noTone(buzzer);
```

This function stops the buzzer sound.

15.Final Delay

```
delay(dt);
```

This adds a 500 ms delay before repeating the loop cycle.

- Conclusion: -

This project successfully demonstrates a motion detection alarm system using Arduino and a PIR sensor. The PIR sensor detects movement and activates both LED and buzzer alarms. The project demonstrates basic IoT and embedded systems concepts through simulation.

Full code Snippet: -

```
int pirPin = 4;
int ledPin = 5;
int buzzer = 6;
int dt=500;

void setup() {
  pinMode(pirPin, INPUT);
  pinMode(ledPin, OUTPUT);
  pinMode(buzzer, OUTPUT);

  Serial.begin(9600);
}

void loop() {

  int motion = digitalRead(pirPin);
  if (motion == HIGH) {
    digitalWrite(ledPin, HIGH);
    Serial.println("Motion Detected!");
    for(int freq = 700; freq <= 2000; freq += 5) {

    tone(buzzer, freq);
    delay(2);
}

for(int freq = 2000; freq >= 700; freq -= 5) {

    tone(buzzer, freq);
    delay(2);
}
  }
  else{
    digitalWrite(ledPin, LOW);
    noTone(buzzer);
    Serial.println("No Motion");
  }
  delay(dt);
}
```